

TIRUMALARAO KILARI

AI Agent Developer · Backend Engineer · LLM Systems Builder

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience designing and evaluating production-grade agentic systems — including multi-stage pipelines, multi-turn LLM interactions, and modular backend architectures. Built a fully autonomous RAG agent featuring session state management, prompt injection safeguards, and a RAGAS-style evaluation harness. Experienced in Python, JavaScript, and SQL across live, non-mocked environments. Currently training and evaluating autonomous AI agents; eager to contribute high-density technical feedback to help Outlier build the next generation of LLM agent systems.

TECHNICAL SKILLS

Agent & LLM: LangGraph (agentic workflows), LangChain, Anthropic Claude, OpenAI GPT-4, Google Gemini, Prompt Engineering, System Prompt Design

Backend & APIs: FastAPI, Uvicorn, REST APIs, Pydantic, Python (primary), JavaScript, SQL, Bash

RAG & Retrieval: FAISS, Sentence-Transformers, Cosine Similarity, Chunking Strategies, Embedding Pipelines, PyMuPDF

Infrastructure: GitHub Actions (CI/CD), Git, GitHub, GitHub Copilot, httpx, python-dotenv

Security & Eval: Prompt injection detection, authority escalation identification, RAGAS-style eval harness, faithfulness scoring

WORK EXPERIENCE

AI Agent & Prompt Engineer (Contract) | **Blue Origin IT Staffing — Client: Austinite Market** · 2024 – Present · Austin, TX

- Designed and deployed a **multi-stage** AI content pipeline: client brief ingestion → prompt engineering → image generation → iterative refinement → production delivery — a real-world **agentic** coordination workflow
- Engineered detailed system prompts in Google Gemini to **autonomously** produce brand-consistent marketing assets; iterated across multiple rounds adjusting style, composition, and output quality
- Delivered production-ready assets physically deployed at the client storefront — validating real-world AI output quality in a live, non-mocked environment
- Demonstrated ability to translate complex business requirements into precise AI **agent** instructions — core to training LLMs for **autonomous**, multi-step task execution

PROJECTS

RAG PDF Chat Agent · Python · FAISS · Claude · FastAPI · LangGraph · pytest · github.com/bizygoodz-web/rag-pdf-chat

- Built a **production-grade autonomous RAG agent** with full **modular** separation: distinct services for PDF ingestion, vector indexing, semantic retrieval, and **Claude**-powered response generation — exactly matching Outlier's requirement for **modular** architecture
- Implemented **multi-turn session state** via persistent **FAISS** index saving/reloading and **MEMORY.md**-style context tracking — enabling the **agent** to maintain state across interactions without re-ingesting documents
- Engineered a zero-hallucination system prompt with explicit **prompt injection** defenses — **agent** refuses out-of-context requests, preventing **authority escalation** and **privacy leaks** from adversarial inputs
- Shipped a **FastAPI** REST server with /ingest, /query, /health, and /sources endpoints operating in a live, non-mocked environment; wrote 10 unit tests (all passing) with **GitHub Actions CI/CD** running on every push

- Built a **RAGAS**-style evaluation harness measuring citation rate, faithfulness, chunk relevance, and latency — providing high-density technical feedback on complex **agent** system behavior

ResumeAI — Autonomous Job Application Agent · Python · FastAPI · Claude · Playwright · github.com/bizygoodz-web/resume-ai

- Designed a **multi-stage autonomous agent** pipeline: job URL ingestion → JD parsing → skill gap analysis → resume rewriting → ATS scoring → auto-application via Playwright browser automation
- Built **modular** backend with distinct services for document parsing, LLM reasoning, and form submission — demonstrating **production-grade** separation of concerns across a complex, multi-step **agent** workflow
- Integrated live external tools including job board APIs and browser automation against real websites — no mocked environments; **agent** handles dynamic forms, session management, and **multi-turn** interactions

EDUCATION

Computer Science | [Lindsey Wilson College](#) · 2022 – 2023 · Columbia, KY

- Independently built **RAG** systems, LLM pipelines, and **agentic** workflows beyond coursework; self-directed study in **LangGraph**, **RAGAS** evaluation, and Anthropic prompt engineering best practices

CERTIFICATIONS & CONTINUOUS LEARNING

- DeepLearning.AI — **LangChain** & Vector Databases in Production (2025, in progress)
- Daily use of GitHub Copilot for AI-assisted development, code review, and **agentic** workflow design
- Deep hands-on experience with **Claude** API (claude-sonnet-4) — system prompts, token management, grounding strategies, and identifying subtle LLM failure modes including **prompt injection** and hallucination
- Studying: **LangGraph** **agentic** workflows, persistent **agent** state management, and multi-**agent** coordination patterns

Open to remote and US-based roles · Available immediately · github.com/bizygoodz-web